

SHREYAS KARNAD

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PROFESSIONAL SUMMARY

Health Informatics graduate of the University of Michigan (MHI, 3.936 GPA) and Biomedical Engineer who pairs deep technical skill in machine learning and healthcare data analytics with the ability to translate that work into decisions non-technical stakeholders can act on. Built predictive models, data pipelines, and dashboards spanning 546,000+ patient records, and consistently reframed technical findings, model results, data quality issues, budget trade-offs, into clear business narratives for leadership, clinicians, and cross-functional teams. Comfortable owning a project end to end: designing the technical solution, building the business case around it, and presenting the recommendation.

EDUCATION

University of Michigan – School of Information, Ann Arbor, MI Aug 2024 – May 2026

Master of Health Informatics (Machine Learning, Data Analytics) - GPA: 3.936

Focus: health data analytics, clinical decision support, FHIR interoperability, AI/ML in healthcare, data governance

Manipal Institute of Technology, MAHE, Manipal, India Oct 2021 – Aug 2023

M.Tech, Biomedical Engineering (Biomedical Instrumentation) - GPA: 3.87

MIT School of Bioengineering Sciences & Research, MIT ADT, Pune, India Aug 2017 – Aug 2021

B.Tech, Bioengineering (Biomedical Engineering elective) - GPA: 3.93

EXPERIENCE

Northwell Health Equity Partners, New York, NY Jul 2026 – Present

Administrative Intern

- Filtered a partner/investor contact database from 12,000+ entries to 305 qualified prospects and translated the findings into a clear recommendation deck; both the methodology and the business takeaways were approved by leadership.
- Built a company tracker for prospective investment targets, integrating financial data (valuation, funding stage, capital raised) with an AI-driven news monitoring system that flags material company developments in plain, actionable terms for the investment team; called "exactly what I wanted" by supervisor.
- Built a structured deal pipeline tracking system, extracting and organizing key deal attributes (sourcing, therapeutic categorization, stage, priority, and follow-up status) from confidential source materials using AI-assisted data extraction.
- Independently designed a persistent database architecture for the firm's AI research agent and presented the technical trade-offs in business terms to VP-level leadership and the engineering lead, securing approval and integration into the firm's production system.
- Shipped a production code change via GitHub pull request updating the agent's tracked-company roster, following a full branch/test/review workflow (baseline tests, feature branch, test suite, code review) under the team's contributor access model.

University of Michigan – School of Public Health, Ann Arbor, MI Jan 2026 – May 2026

Grader, Public Health Informatics (Part-time)

- Evaluated 98+ graduate-level assignments per cycle using structured rubrics, maintaining consistent scoring standards and reducing instructor review time through detailed written feedback on analytical approach and data interpretation.
- Collaborated directly with Prof. Rahul Ladhania on grading rubric design, dispute resolution, and quality assurance across multiple assignment cycles; identified and corrected systematic grading inconsistencies before grade release.

University of Michigan – School of Information, Ann Arbor, MI Sep 2025 – Dec 2025

Instructional Aid, Health Informatics / Database Management (Part-time)

- Supported a 150+ student health informatics course by grading analytical examinations, proctoring assessments, and communicating complex technical concepts clearly to students with varying programming backgrounds.
- Collaborated directly with Prof. Bobby Madamanchi on exam paper design, dispute resolution, and quality assurance across 3 exam cycles; identified and corrected systematic grading inconsistencies before grade release.

Biocomplexity Institute, Indiana University, Bloomington, IN (Remote) Jun 2025 – Aug 2025

Data Science Intern

- Designed schema-based JSON pipelines to automatically generate syntactically valid CompuCell3D biological simulation code from research paper abstracts; documented zero-shot LLM failure modes (version mismatches, hallucinated APIs, incorrect biological parameters) to guide iterative pipeline refinement.
- Translated natural-language biological modeling questions into structured computational parameters (adhesion, chemotaxis, diffusion) via an NL → ontology → code framework, enabling non-programmer researchers to generate simulation-ready outputs.
- Re-engineered a 9-agent CrewAI system (Scientist, Team Leader, Coder, Ontology Expert, Biological Modeling Expert, Version Checker, Coverage Verification, Packager, Interpreter) to replace OpenAI with DeepSeek Replicate via a custom Python wrapper in VS Code, resolving API environment conflicts and reducing inference costs while maintaining code

generation accuracy.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India (Remote)

Aug 2023 – Aug 2024

Associate Biomedical Engineer

- Evaluated deep learning classification models (ResNet50, GoogLeNet, DenseNet) in MATLAB Deep Learning Toolbox to classify harmonic signature data from magnetic nanoparticle-based intraoperative sentinel lymph node detection; achieved 96–100% accuracy distinguishing tumor-present from tumor-absent nodes, and communicated results to clinical stakeholders in plain, decision-relevant terms to support adoption.
- Integrated MATLAB with LabVIEW to interface with MafPro magnetic field probe hardware; deployed ML inference pipelines to AWS S3-connected cloud servers, enabling remote clinics without high-end NPU hardware to offload classification tasks to cloud infrastructure and receive results in real time.
- Designed a ruggedized field enclosure in Fusion 360 housing a Surface Pro with NPU, battery pack, and CAT8 magnetic detector port; implemented a locked-down Windows 11 environment restricting end-user access to MafPro software only, producing a deployment-ready clinical device prototype.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India

Aug 2022 – Jul 2023

Graduate Research Intern

- Interviewed 20+ surgical oncologists to translate their clinical workflow needs into formal product requirements, bridging the gap between clinical end users and the technical development roadmap.
- Built and validated a magnetic probe prototype for intraoperative sentinel lymph node detection; analyzed experimental sensor data using time-series methods and contributed findings to M.Tech thesis and internal technical reporting.

TATA Elxsi, St. Paul, MN (Remote)

Jan 2021 – Jun 2021

Project Intern

- Analyzed investment data across 50+ digital health companies totaling over \$12B and translated the findings into clear sector rankings and a plain-language recommendation memo used directly by leadership to inform strategic investment decisions.
- Mapped Software as a Medical Device (SaMD) regulatory frameworks across FDA and IMDRF pathways, producing a risk classification reference guide used to inform product development planning.

PROJECTS

MIMIC-IV Health Equity & Disparities Intelligence Dashboard

Apr 2026 – Jun 2026

- Architected a production-realistic Power BI star schema on 546,028 MIMIC-IV v3.1 hospitalizations (223,452 patients) using Import Mode with pre-aggregated BigQuery tables rather than live DirectQuery, reducing ongoing cloud cost to zero while preserving full analytical granularity across 7 race/ethnicity, 3 insurance, and 4 age dimensions.
- Built 6 BigQuery aggregation objects (1 staging view + 5 fact tables) and a star schema data model with 4 dimension tables to resolve Many-to-Many relationship ambiguity; authored 21 DAX measures using SUMX/DIVIDE population-weighted patterns across volume, length-of-stay, disparity-gap, and condition-level categories.
- Diagnosed and corrected a data quality bug producing a mathematically impossible 289% KPI value, tracing the root cause to incorrect aggregation of pre-computed percentage columns across non-uniform denominator populations; re-derived all affected measures from underlying admission counts and proactively applied the same fix to two additional at-risk measures.
- Built a 6-page, 40-visual interactive dashboard and translated the findings into a single executive narrative: a >2x stroke mortality gap (31.09% Unknown/Declined vs. 14.03% White) and a ~3.2x Acute MI mortality gap (American Indian/Alaska Native vs. White), framed for hospital leadership decision-making across 10 ICD-10-derived clinical condition groups.

MIMIC-IV Pre-Treatment Sepsis Early Warning System

Apr 2026 – Jun 2026

- Motivated by Wiens et al.'s (NEJM AI, 2024) finding that the Epic Sepsis Model's AUROC collapses from 0.62 to 0.47 on Michigan Medicine data when restricted to pre-treatment predictions, trained a Random Forest classifier on 67,286 PhysioNet-credentialed MIMIC-IV v3.1 adult ICU stays, achieving AUROC 0.7766–0.8160, a +0.31 to +0.35 improvement over the published ESM pre-treatment baseline.
- Conducted a SHAP TreeExplainer contamination audit across all 76 features confirming zero treatment markers influenced predictions; generated a FHIR R4 RiskAssessment interoperability layer with a custom pre-treatment certification extension and SNOMED CT coding for Epic EHR integration via SMART on FHIR.
- Engineered 8 BigQuery SQL extraction queries across MIMIC-IV hospital, ICU, and derived schemas; built a Power BI Desktop dashboard with DirectQuery to BigQuery, DAX measures, calculated columns, and calculated tables; validated full pipeline reproducibility across two compute environments (Colab AUROC 0.7766; VSCode AUROC 0.8160). GitHub: github.com/karnadsp/mimic-sepsis-ews

APP-2-MARKET Product Launch Strategy – SI 661, University of Michigan

Jan 2026 – Apr 2026

- Led project management and business planning for a health data app's go-to-market strategy, structuring the work into a 6-phase, 30+ task Work Breakdown Structure (competitive strategy, maintenance, distribution/compliance, data security, testing, and financial planning).
- Built a full project budget spanning labor, materials, and fixed costs (\$92,000 projected vs. \$93,150 actual, a 1.25% variance), and made and defended sequencing trade-offs, prioritizing compliance work ahead of broader rollout, to stakeholders under live Q&A.

- Delivered a comprehensive executive summary and financial model for a hypothetical health-app company (Tradarbol Tech), translating technical product requirements (HL7 FHIR compliance, data security architecture) into business terms for funding and partnership decisions.

EHR AI Hallucination Mitigation Strategy

Mar 2026 – Apr 2026

- Identified that AI-generated hallucinations in EHR transcription tools create undetected clinical documentation errors; proposed a Zero-Trust RAG framework with retrieval-based validation checkpoints, reducing reliance on unverified LLM outputs and improving documentation accuracy and HIPAA compliance.

Ctrl Alt Drink - Mobile Health Intervention for Student Alcohol Moderation

Jan 2026 – Apr 2026

- Designed a behavioral health app targeting student alcohol moderation; incorporated real-time behavioral tracking, anonymous peer support, and micro-goal frameworks grounded in evidence-based behavior change principles to address a gap in accessible, stigma-free campus health tools.

Structural Health Disparities and Sepsis Risk Analysis

Sept 2025 – Dec 2025

- Analyzed a large administrative California county-level healthcare dataset to determine structural drivers of sepsis risk; built a Random Forest classification model achieving 80% accuracy and multivariate regression identifying insurance coverage as the primary predictor of postoperative outcomes.
- Authored full statistical methods and results sections for academic submission, including data quality validation, feature engineering, and county-level health disparity visualizations.

Strategic Evaluation of Omega Hospital Orthopedic & Cardiology Service Lines – HMP669, University of Michigan

Mar 2025 – Apr 2025

- Led data cleaning and export within Microsoft Access on a large administrative hospital database (.accdb), resolving Access-specific compatibility issues across team members' operating systems; standardized inconsistent records to ensure data integrity before analysis.
- Designed the project's risk assessment framework (severity/likelihood risk matrix) used to evaluate carve-out contract risks across cardiology and orthopedic service lines.
- Collaborated with a 3-person team to analyze patient demographics, financial performance, and resource utilization using RStudio, finding cardiology generated \$910 average profit per case versus a \$470 average loss per case in orthopedics; co-authored a written report recommending cardiology carve-out contracts while deferring orthopedic contracts pending cost-driver analysis.

Python-Based Diabetes Risk Assessment App + FHIR Output

Oct 2024 – Nov 2024

- Built an end-to-end ML pipeline for diabetes risk prediction; generated structured FHIR-formatted outputs to enable interoperable data sharing across health systems, demonstrating real-world applicability of clinical decision support tools.

Diabetes Risk Analysis - SI 544, University of Michigan

Sep 2024 – Dec 2024

- Applied hypothesis testing (two-sample t-tests) and multivariate linear regression in R to a 768-patient Pima Indians diabetes dataset; identified blood glucose ($p < 2.2e-16$) and BMI as statistically significant primary predictors while surfacing zero-value missingness as a systematic data quality issue affecting model reliability.

Breast Cancer Tumor Network Analysis - SI 507, University of Michigan

Feb 2025 – Apr 2025

- Compared K-Means and Spectral Clustering algorithms on the Wisconsin Diagnostic breast cancer dataset (32 features, 569 patients); Spectral Clustering outperformed on accuracy and runtime, while network centrality analysis identified `concave_points_mean` as the most diagnostically influential tumor feature across degree, closeness, and betweenness metrics.

IVF Success Prediction - Binary Classification, Manipal Institute of Technology

April 2022 – June 2022

- Applied Logistic Regression, K-Nearest Neighbour, and Naïve Bayes classifiers in Python to a clinical IVF dataset (55 patients, 14 metabolites); KNN achieved best accuracy of 83% on optimized metabolite combinations; identified data volume as primary constraint on model reliability.

PPI2 Web Server - Protein-Protein Interaction Database

Jan 2019 – Dec 2020

- Developed an HTML-based research data system interfacing with a relational biological database; integrated interactive 3D protein interaction visualizations to support researcher queries and data exploration.

PAPERS

Current Trends of AI and Applications in Digital Pathology: A Comprehensive Review

Dec 2023

- *International Journal of Advanced and Applied Sciences (IJAAS)* - peer-reviewed publication

TECHNICAL SKILLS

Business & Communication: stakeholder translation, executive reporting, project budgeting, work breakdown structures, risk assessment | **Programming:** Python (pandas, scikit-learn, matplotlib, seaborn), R (tidyverse, ggplot2, caret), SQL, Java, C/C++ (Arduino/embedded), MATLAB, LabVIEW | **GenAI/LLM:** CrewAI, OpenAI APIs, DeepSeek Replicate, Claude (Opus), prompt engineering, multi-agent workflows, RAG, ontology extraction, AI-assisted data extraction | **ML/Stats:** predictive modeling, classification, regression, clustering, hypothesis testing, biostatistics, time-series analysis, harmonic signal analysis | **Deep Learning:** ResNet50, GoogLeNet, DenseNet, MATLAB Deep Learning Toolbox, image classification, biological

signal classification | **Healthcare/Interop:** FHIR (file generation), HL7, HIE architecture, EHR systems, clinical decision support, HIPAA, 42 CFR | **Data:** MySQL, Microsoft Access, Excel; data pipeline design, EDA, feature engineering, data quality assurance | **Cloud/Tools:** AWS (S3, server-based inference), Google BigQuery, Git, VS Code, Fusion 360; technical documentation, reproducible workflows | **BI & Visualization:** Power BI Desktop (Import Mode, DirectQuery, DAX measures, star schema modeling, calculated columns/tables), SHAP TreeExplainer, matplotlib, seaborn | **Database/DevOps:** SQLite, database schema design, Git-based contributor workflow (branch/PR/code review), production system integration